



CAT 2025 Slot 2 Question Paper - QA

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Quant

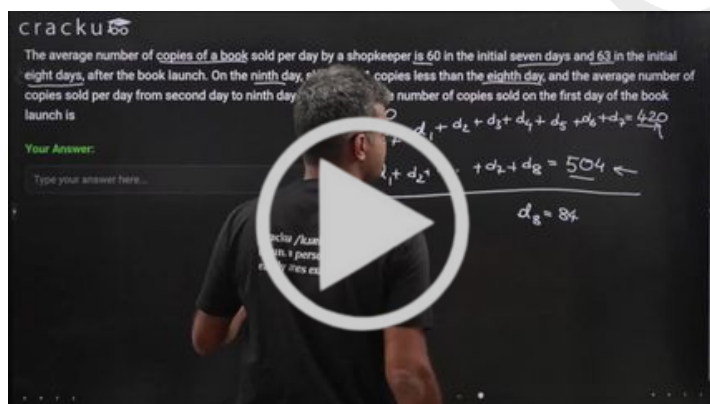
1. If $9x^2 + 2x - 3 - 4(3x^2 + 2x - 2) + 27 = 0$ then the product of all possible values of x is

- A 30
- B 20
- C 5
- D 15



VIDEO SOLUTION

2. The average number of copies of a book sold per day by a shopkeeper is 60 in the initial seven days and 63 in the initial eight days, after the book launch. On the ninth day, she sells 11 copies less than the eighth day, and the average number of copies sold per day from second day to ninth day becomes 66. The number of copies sold on the first day of the book launch is



VIDEO SOLUTION

3. The set of all real values of x for which $(x^2 - |x + 9| + x) > 0$, is

- A $(-\infty, -3) \cup (3, \infty)$
- B $(-\infty, -9) \cup (3, \infty)$

C $(-9, -3) \cup (3, \infty)$

D $(-\infty, -9) \cup (9, \infty)$



VIDEO SOLUTION

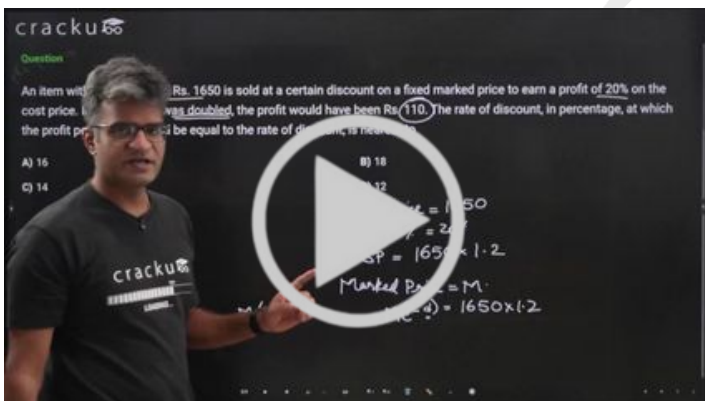
4. An item with a cost price of Rs. 1650 is sold at a certain discount on a fixed marked price to earn a profit of 20% on the cost price. If the discount was doubled, the profit would have been Rs. 110. The rate of discount, in percentage, at which the profit percentage would be equal to the rate of discount, is nearest to

A 16

B 18

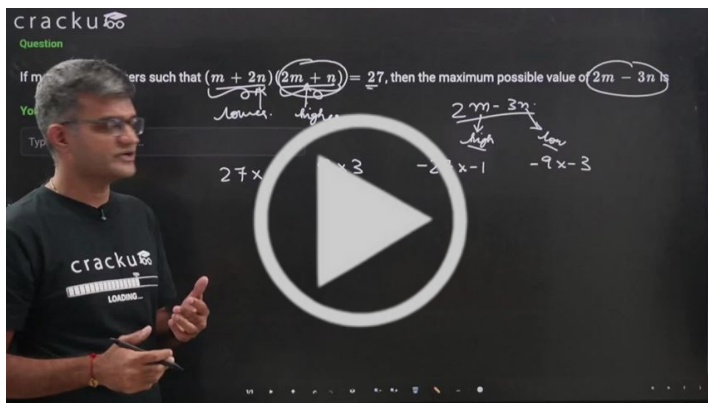
C 14

D 12



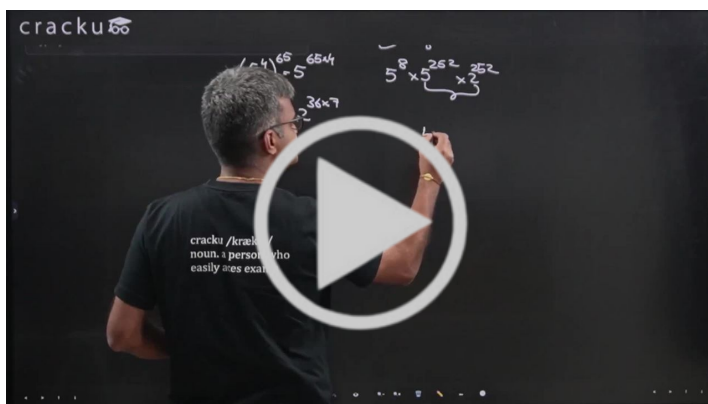
VIDEO SOLUTION

5. If m and n are integers such that $(m + 2n)(2m + n) = 27$, then the maximum possible value of $2m - 3n$ is



VIDEO SOLUTION

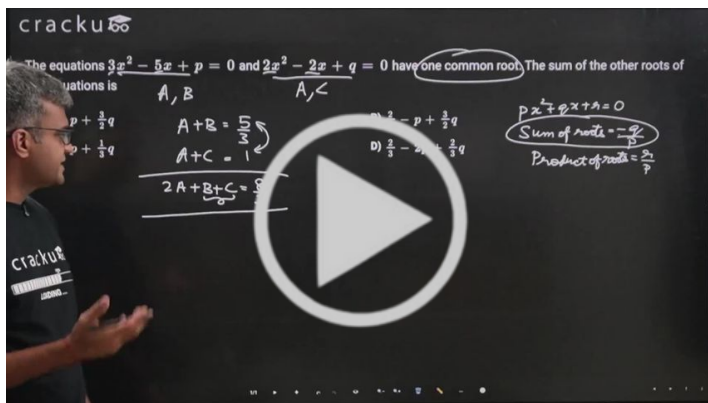
6. The sum of digits of the number $(625)^{65} \times (128)^{36}$ is



VIDEO SOLUTION

7. The equations $3x^2 - 5x + p = 0$ and $2x^2 - 2x + q = 0$ have one common root. The sum of the other roots of these equations is

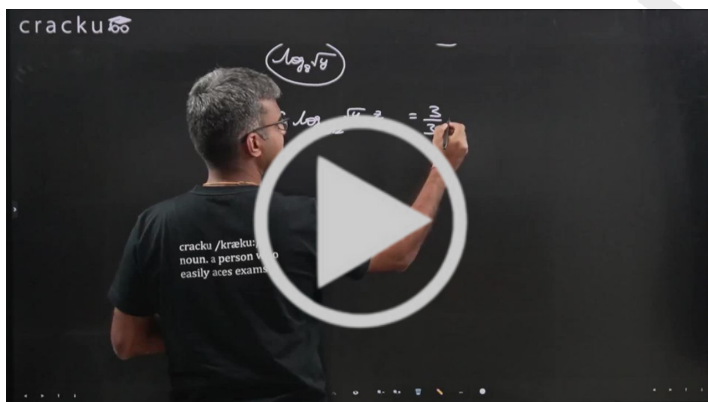
- A $\frac{8}{3} - p + \frac{3}{2}q$
- B $\frac{2}{3} - p + \frac{3}{2}q$
- C $\frac{8}{3} + p + \frac{1}{3}q$
- D $\frac{2}{3} - 2p + \frac{2}{3}q$



VIDEO SOLUTION

8. If $\log_{64} x^2 + \log_8 \sqrt{y} + 3 \log_{512} (\sqrt{y}z) = 4$, where x, y and z are positive real numbers, then the minimum possible value of $(x + y + z)$ is

- A 48
- B 36
- C 24
- D 96



VIDEO SOLUTION

9. Rita and Sneha can row a boat at 5 km/h and 6 km/h in still water, respectively. In a river flowing with a constant velocity, Sneha takes 48 minutes more to row 14 km upstream than to row the same distance downstream. If Rita starts from a certain location in the river, and returns downstream to the same location, taking a total of 100 minutes, then the total distance, in km, Rita will cover is



VIDEO SOLUTION

10. Suppose a, b, c are three distinct natural numbers, such that $3ac = 8(a + b)$. Then, the smallest possible value of $3a + 2b + c$ is



VIDEO SOLUTION

11. Let $f(x) = \frac{x}{(2x-1)}$ and $g(x) = \frac{x}{(x-1)}$. Then the domain of the function $h(x) = f(g(x)) + g(f(x))$ is all real numbers except

- A $-1, \frac{1}{2}, \text{ and } 1$
- B $\frac{1}{2}, 1, \text{ and } \frac{3}{2}$
- C $-\frac{1}{2}, \frac{1}{2}, \text{ and } 1$
- D $\frac{1}{2}, \text{ and } 1$



- A** 10
B 11
C 9
D 8

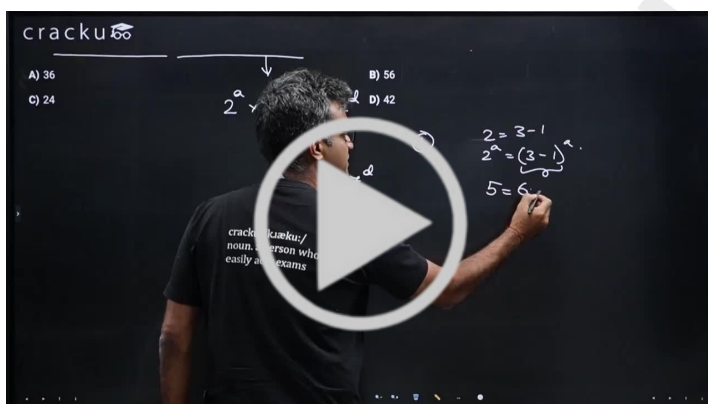




VIDEO SOLUTION

14. The number of divisors of $(2^6 \times 3^5 \times 5^3 \times 7^2)$, which are of the form $(3r + 1)$, where r is a non-negative integer, is

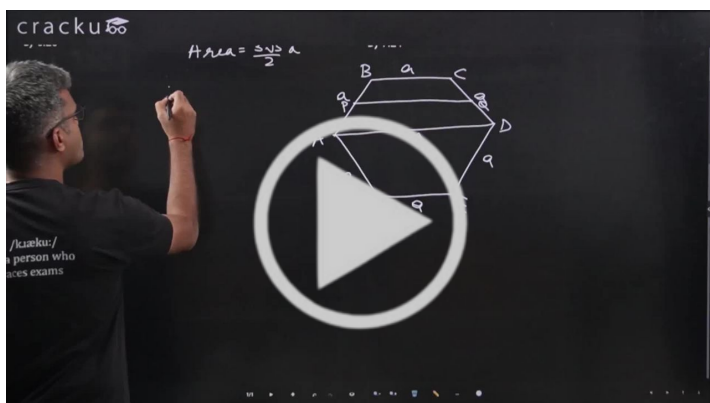
- A 36
- B 56
- C 24
- D 42



VIDEO SOLUTION

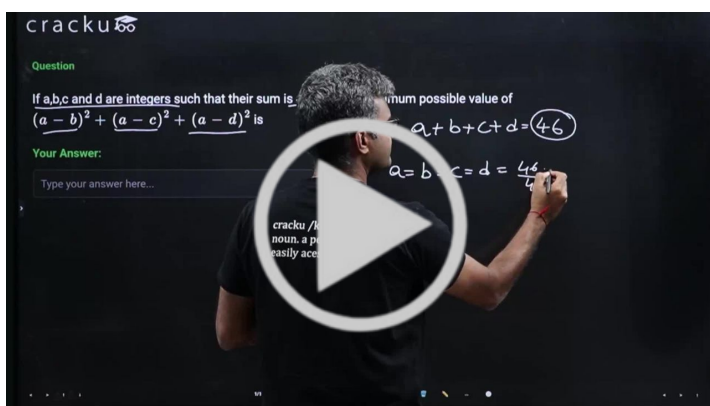
15. Let ABCDEF be a regular hexagon and P and Q be the midpoints of AB and CD, respectively. Then, the ratio of the areas of trapezium PBCQ and hexagon ABCDEF is

- A 6:19
- B 5:24
- C 6:25
- D 7:24



VIDEO SOLUTION

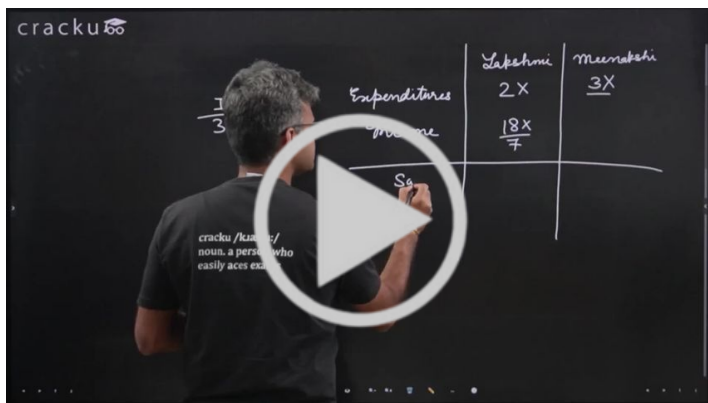
16. If a, b, c and d are integers such that their sum is 46, then the minimum possible value of $(a - b)^2 + (a - c)^2 + (a - d)^2$ is



VIDEO SOLUTION

17. The ratio of expenditures of Lakshmi and Meenakshi is 2 : 3, and the ratio of income of Lakshmi to expenditure of Meenakshi is 6 : 7. If excess of income over expenditure is saved by Lakshmi and Meenakshi, and the ratio of their savings is 4 : 9, then the ratio of their incomes is

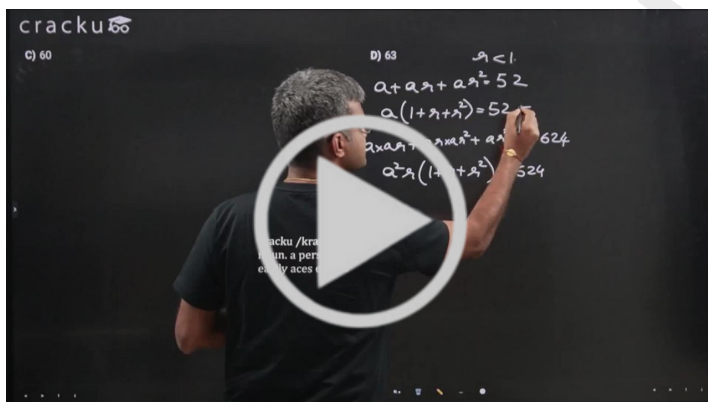
- A 3:5
- B 5:6
- C 2:1
- D 7:8



VIDEO SOLUTION

18. Let a_n be the n^{th} term of a decreasing infinite geometric progression. If $a_1 + a_2 + a_3 = 52$ and $a_1a_2 + a_2a_3 + a_3a_1 = 624$, then the sum of this geometric progression is

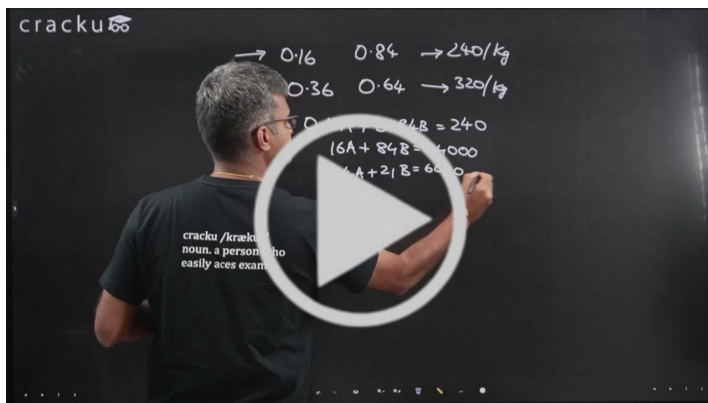
- A 57
- B 54
- C 60
- D 63



VIDEO SOLUTION

19. A mixture of coffee and cocoa, 16% of which is coffee, costs Rs 240 per kg. Another mixture of coffee and cocoa, of which 36% is coffee, costs Rs 320 per kg. If a new mixture of coffee and cocoa costs Rs 376 per kg, then the quantity, in kg, of coffee in 10 kg of this new mixture is

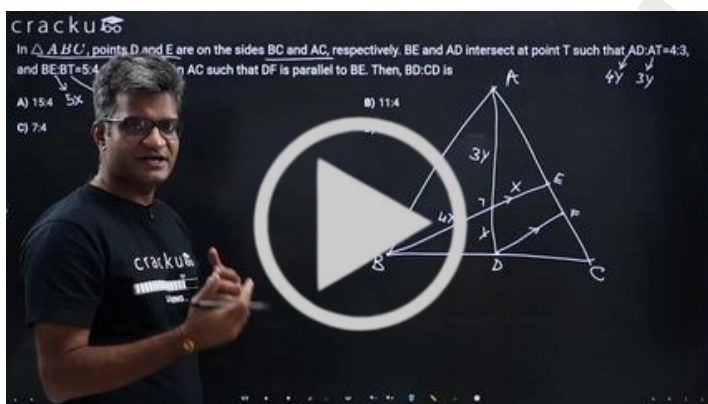
- A 5
- B 4
- C 2.5
- D 6



VIDEO SOLUTION

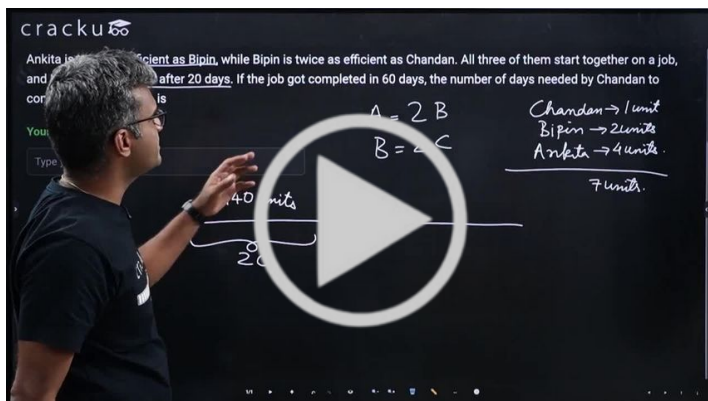
20. In $\triangle ABC$, points D and E are on the sides BC and AC, respectively. BE and AD intersect at point T such that $AD:AT=4:3$, and $BE:BT=5:4$. Point F lies on AC such that DF is parallel to BE. Then, $BD:CD$ is

- A 15:4
- B 11:4
- C 7:4
- D 9:4



VIDEO SOLUTION

21. Ankita is twice as efficient as Bipin, while Bipin is twice as efficient as Chandan. All three of them start together on a job, and Bipin leaves the job after 20 days. If the job got completed in 60 days, the number of days needed by Chandan to complete the job alone, is



[VIDEO SOLUTION](#)

22. A certain amount of money was divided among Pinu, Meena, Rinu and Seema. Pinu received 20% of the total amount and Meena received 40% of the remaining amount. If Seema received 20% less than Pinu, the ratio of the amounts received by Pinu and Rinu is

- A 2:1
- B 1:2
- C 5:8
- D 8:5



[VIDEO SOLUTION](#)

Answers

Quant

1.B	2.49	3.A	4.C	5.17	6.25	7.A	8.A
9.8	10.12	11.A	12.D	13.80	14.D	15.B	16.2
17.A	18.B	19.A	20.B	21.340	22.C		

Cracku

Explanations

Quant

1. **B**

Let's assume that $x^2 + 2x - 3 = t$

$9^{x^2+2x-3} - 4(3^{x^2+2x-2}) + 27 = 0$ can be written as $9^t - 4(3^{t+1}) + 27 = 0$

$$3^{2t} - 12(3^t) + 27 = 0$$

Let's assume that $3^t = y$

$$y^2 - 12y + 27 = 0$$

$$(y - 9)(y - 3) = 0$$

$$y = 3 \text{ or } 9 \Rightarrow t = 1 \text{ or } 2$$

Let's solve when $t = 1$

$$x^2 + 2x - 3 = 1 \Rightarrow x^2 + 2x - 4 = 0$$

$$b^2 - 4ac = 4 + 16 = 20$$

Positive, so the equation has real roots.

Product of possible value of $x = -4$

Let's solve for $t = 2$

$$x^2 + 2x - 3 = 2 \Rightarrow x^2 + 2x - 5 = 0$$

$$b^2 - 4ac = 4 + 20 = 24$$

Positive, so the equation has real roots.

Product of possible value of $x = -5$

The product of all values = 20

 VIDEO SOLUTION

2. **49**

Let the copies sold on days 1 through 9 be $d_1, \dots, d_9$

From the averages:

$$d_1 + \dots + d_7 = 7 \times 60 = 420$$

$$d_1 + \dots + d_8 = 8 \times 63 = 504$$

$$\text{So } d_8 = 504 - 420 = 84. \text{ Then } d_9 = 84 - 11 = 73$$

The sum from day 2 to day 9 is $8 \times 66 = 528$

$$d_2 + \dots + d_9 = (d_1 + \dots + d_8) - d_1 + d_9 = 504 - d_1 + 73 = 577 - d_1$$

$$\text{Thus } 577 - d_1 = 528 \Rightarrow d_1 = 49$$

 VIDEO SOLUTION

3. **A**

We are asked to solve

$$x^2 - |x + 9| + x > 0$$

Split into two cases based on the absolute value.

Case 1: $x + 9 \geq 0 \Rightarrow x \geq -9$

$$|x + 9| = x + 9$$

Inequality becomes: $x^2 - (x + 9) + x > 0 \Rightarrow x^2 - 9 > 0 \Rightarrow (x - 3)(x + 3) > 0$

So $x < -3$ or $x > 3$. Combined with $x \geq -9$, we get $x \in [-9, -3) \cup (3, \infty)$

Case 2: $x + 9 < 0 \Rightarrow x < -9$

$$|x + 9| = -(x + 9) = -x - 9$$

Inequality becomes: $x^2 - (-x - 9) + x > 0 \Rightarrow x^2 + 2x + 9 > 0$

The quadratic $x^2 + 2x + 9$ has discriminant $(4 - 36 = -32 < 0)$, so always positive. But in this case ($x < -9$), so inequality is satisfied. Thus ($x < -9$) also works.

$$x < -3 \text{ or } x > 3$$

So the solution set is $(-\infty, -3) \cup (3, \infty)$

 VIDEO SOLUTION

4. C

Let the marked price be M and the initial discount rate be d . The cost price is 1650. A profit of 20% means the selling price is

$$SP = 1650 \times 1.2 = 1980$$

With discount d ,

$$M(1 - d) = 1980$$

If the discount is doubled, the selling price becomes $(M(1 - 2d))$, and the profit is 110, so

$$M(1 - 2d) - 1650 = 110 \Rightarrow M(1 - 2d) = 1760$$

Subtracting the two equations:

$$M(1 - d) - M(1 - 2d) = 1980 - 1760 \Rightarrow Md = 220 \Rightarrow M = \frac{220}{d}$$

Plug into the first equation:

$$\frac{220}{d}(1 - d) = 1980 \Rightarrow \frac{1 - d}{d} = 9 \Rightarrow 1 - d = 9d \Rightarrow d = 0.1$$

So the initial discount rate is 10% and the marked price is

$$M = \frac{220}{0.1} = 2200$$

Now, let the discount rate be (r) such that the profit percentage equals the discount percentage. Then

$$\frac{2200(1 - r) - 1650}{1650} = r$$

Simplify:

$$2200(1 - r) - 1650 = 1650r \Rightarrow 2200 - 2200r - 1650 = 1650r \Rightarrow 550 = 3850r \Rightarrow r = \frac{550}{3850} \approx 0.142857$$

Thus, the required discount rate is approximately 14%

 VIDEO SOLUTION

5. 17

Sure! We can solve it without explicitly introducing new variables for substitution.

We are given $(m + 2n)(2m + n) = 27$ and we want to maximize $2m - 3n$

Because m and n are integers, both $(m + 2n)$ and $(2m + n)$ are going to be integers.

Factor pairs of 27 (including negatives) are:

$(1, 27), (3, 9), (9, 3), (27, 1), (-1, -27), (-3, -9), (-9, -3), (-27, -1)$

For each factor pair (a,b) , take $a = m + 2n, b = 2m + n$ and solve for integers (m,n) .

If we notice $a + b = (m + 2n) + (2m + n) = 3(m + n)$

So, basically the sum of the two numbers has to be a multiple of 3.

So, the ordered pairs we will consider are $(3, 9), (9, 3), (-3, -9), (-9, -3)$

For $(a, b) = (9, 3)$:

$$m + 2n = 9, \quad 2m + n = 3$$

Upon solving, we get $n = 5$ and $m = -1$

For $(a, b) = (3, 9)$:

$$m + 2n = 3, \quad 2m + n = 9$$

Upon solving, we get $n = -1$ and $m = 5$

Then $2m - 3n = 2 \cdot 5 - 3(-1) = 10 + 3 = 13$

For negative factor pairs, we similarly get integer solutions $(-5, 1)$ and $(1, -5)$, giving $2m - 3n = -13$ and 17 , respectively.

Thus, the maximum value of $(2m - 3n)$ among all solutions is 17

 VIDEO SOLUTION

6.25

$$625 = 5^4 \text{ and } 128 = 2^7$$

$$\text{So, } (625)^{65}, (128)^{36} = 5^{260}, 2^{252} = (2^{252} 5^{252}) \cdot 5^8 = 10^{252} \cdot 5^8$$

Thus, the number equals 5^8 followed by 252 zeros. Now $(5^8=390625)$, and the sum of the digits of (390625) is

$$3 + 9 + 0 + 6 + 2 + 5 = 25.$$

Zeros add nothing, so the required sum of digits is 25.

 VIDEO SOLUTION

7.A

Let's assume that the common root is r .

The sum of the roots of the first equation is $5/3$ and that of the second equation is 1.

We want the sum of the other two roots:

$$\text{Sum} = \left(\frac{5}{3} - r\right) + (1 - r) = \frac{8}{3} - 2r$$

We now need to express r in terms of p and q .

Since r is a common root, it satisfies:

$$3r^2 - 5r + p = 0 \quad (1)$$

$$2r^2 - 2r + q = 0 \quad (2)$$

Eliminate r^2 .

Multiply (2) by 3:

$$6r^2 - 6r + 3q = 0$$

Multiply (1) by 2:

$$6r^2 - 10r + 2p = 0$$

Subtract:

$$(6r^2 - 6r + 3q) - (6r^2 - 10r + 2p) = 0$$

$$4r + 3q - 2p = 0$$

$$r = \frac{2p-3q}{4}$$

Now substitute into $\frac{8}{3} - 2r$:

$$\frac{8}{3} - 2\left(\frac{2p-3q}{4}\right)$$

$$= \frac{8}{3} - \frac{2p-3q}{2}$$

$$= \frac{8}{3} - p + \frac{3}{2}q$$

 VIDEO SOLUTION

8. A

$$64 = 8^2 \text{ and } 512 = 8^3$$

$$\log_{64} x^2 + \log_8 \sqrt{y} + 3 \log_{512} (\sqrt{y}z) = 4,$$

Property of log: $\log_{b^m} a^n = \frac{n}{m} \log_b a$

$$\log_{8^2} x^2 + \log_8 \sqrt{y} + 3 \log_{8^3} (\sqrt{y}z) = 4$$

Using the above-mentioned property, the expression becomes $\log_8 x + \log_8 \sqrt{y} + \log_8 (\sqrt{y}z) = 4$

$$\log_8 x \sqrt{y} \cdot (\sqrt{y}z) = 4$$

$$\log_8 xyz = 4$$

$$xyz = 8^4 = 2^{12}$$

Using AM-GM inequality

$$\frac{(x+y+z)}{3} \geq \sqrt[3]{xyz}$$

$$\frac{(x+y+z)}{3} \geq 2^4$$

$$(x + y + z) \geq 48$$

 VIDEO SOLUTION

9. 8

Let the river speed be v km/h. For Sneha still-water speed = 6 km/h:

Upstream speed = $(6-v)$, downstream speed = $(6+v)$.

$$\text{Given } \frac{14}{(6-v)} - \frac{14}{(6+v)} = 48 \text{ minutes} = 0.8 \text{ hours.}$$

$$\text{So, } \frac{14(6+v-6+v)}{36-v^2} = 0.8 \Rightarrow \frac{28v}{36-v^2} = 0.8$$

$$28v = 0.8(36-v^2) = 28.8 - 0.8v^2 \Rightarrow 0.8v^2 + 28v - 28.8 = 0$$

$$\text{Multiply by 5: } 4v^2 + 140v - 144 = 0$$

$$v^2 + 35v - 36 = 0$$

We get $v = 1$

Thus, the river speed is 1 km/h. For Rita, the still water speed is 5 km/h:

Upstream speed = $5-1=4$ km/h, Downstream Speed = $(5+1=6)$ km/h.

If she rows d km upstream and returns d km downstream, the total time

$$\frac{d}{4} + \frac{d}{6} = d\left(\frac{1}{4} + \frac{1}{6}\right) = d \cdot \frac{5}{12} \text{ hours}$$

This equals 100 minutes $= 100/60 = \frac{5}{3}$ hours.

$$\text{So, } d \cdot \frac{5}{12} = \frac{5}{3} \Rightarrow d = 4 \text{ km}$$

Total distance covered $2d=8$ km.

 VIDEO SOLUTION

10.12

Our task is to minimise $3a + 2b + c$.

Here, the coefficient for c is the minimum.

$$3ac = 8(a + b)$$

We know that a , b , and c are natural numbers. So, the product ac should definitely be a multiple of 8.

$$\text{Case 1: } a = 1, c = 8 \text{ and } b = 2 \Rightarrow 3a + 2b + c = 15$$

$$\text{Case 2: } a = 2, c = 4 \text{ and } b = 1 \Rightarrow 3a + 2b + c = 12$$

So, 12 is the correct answer.

 VIDEO SOLUTION

11.A

We check where $f(x) = \frac{x}{2x-1}$ or $g(x) = \frac{x}{x-1}$, or their compositions, become undefined.

First, $f(x)$ is undefined at $x = \frac{1}{2}$ and $g(x)$ is undefined at $x = 1$.

$$\text{Next, for } f(g(x)) = \frac{x}{x+1}$$

Since, the denominator can't be zero, $x=-1$ must also be excluded.

$$\text{For } g(f(x)) = \frac{x}{1-x}$$

$\Rightarrow x = 1$ is not possible, which is already excluded.

So the values at which $h(x) = f(g(x)) + g(f(x))$ is undefined are $-1, \frac{1}{2}, 1$

 VIDEO SOLUTION

12.D

Let the annual interest rate be (r) (in decimal). Discount the two instalments to present value:

$$\frac{530}{1+r} + \frac{594}{(1+r)^2} = 1000$$

Set $x = \frac{1}{1+r}$. Then

$$594x^2 + 530x - 1000 = 0$$

$$\text{Discriminant} = 530^2 + 4 \cdot 594 \cdot 1000 = 280900 + 2376000 = 2656900 = 1630^2.$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-530 + 1630}{2 \cdot 594} = \frac{1100}{1188} = \frac{275}{297}$$

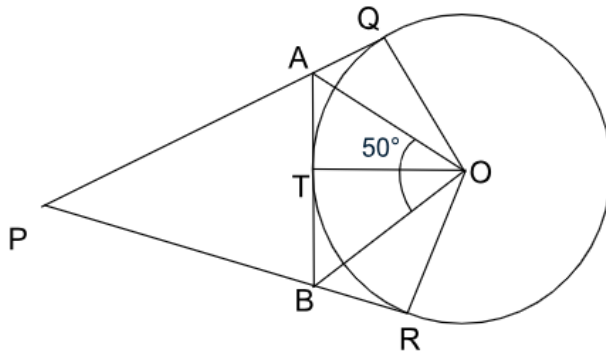
$$\text{So } 1 + r = \frac{297}{275}$$

$$r = \frac{297 - 275}{275} = \frac{22}{275} = \frac{2}{25} = 0.08 = 8\%$$

[VIDEO SOLUTION](#)

13.80

We can draw the following diagram as per the given question :



Now, taking $\triangle AOT$ and $\triangle AOQ$

AO is the common side

AT=AQ (Tangents drawn from an external point are equal in length)

OT=OQ (radius of circle)

So, by S.S.S., $\triangle AOT$ is congruent to $\triangle AOQ$

So, by C.P.C.T., $\angle AOT = \angle AOQ$

Similarly, for $\triangle BOT$ and $\triangle BOR$, we can do the same thing,

So, we can say $\triangle BOT$ and $\triangle BOR$ will also be congruent

Now, $\angle QOR = \angle AOQ + \angle AOB + \angle ROB = \angle AOT + \angle AOB + \angle BOT = 2\angle AOB = 2 \times 50^\circ = 100^\circ$

Now, in quadrilateral QOPR,

$\angle OQP = \angle ORP = 90^\circ$ (Tangents drawn to a point is perpendicular to the radius drawn at the same point)

Now, sum of angles of a quadrilateral is 360°

So, $\angle APB + \angle QOR = 360^\circ - (90^\circ + 90^\circ) = 180^\circ$

So, $\angle APB = 180^\circ - 100^\circ = 80^\circ$

[VIDEO SOLUTION](#)

14.D

The divisors of the given number will have the form $2^a * 3^b * 5^c * 7^d$ with $0 \leq a \leq 6$, $0 \leq b \leq 5$, $0 \leq c \leq 3$, $0 \leq d \leq 2$

Because the divisors should be in the form $3r+1$, it cannot be divisible by 3, so $(b=0)$.

Reduce modulo 3: $2 \bmod 3 = 2$, $5 \bmod 3 = 2$, $7 \bmod 3 = 1$

Hence, $2^a 5^c 7^d \equiv 2^{a+c} \cdot 1^d \equiv 2^{a+c} \pmod{3}$

2^k will be in the form $3r+1$ only when k is even. So, we need $a+c$ to be even

$a \in 0, \dots, 6$ has 4 even, 3 odd values

$c \in 0, \dots, 3$ has 2 even, 2 odd

Number of (a,c) with $(a+c)$ even is $4 \cdot 2 + 3 \cdot 2 = 8 + 6 = 14$

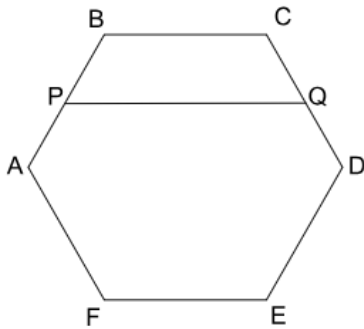
For each such pair, there are 3 choices for $d = 0, 1, 2$. Thus, total divisors in the form $(3r+1)$ equals $14 \times 3 = 42$.

[VIDEO SOLUTION](#)

15. B

Let the side of the hexagon be a .

The area of the whole hexagon is going to be $\frac{3\sqrt{3}a^2}{2}$



We know that the longest diagonal of a hexagon is 2 times the side of the hexagon.

$$PQ = \frac{(BC+AD)}{2} = \frac{(a+2a)}{2} = 1.5a$$

The area of the trapezium = Average of parallel sides * Height = $\frac{(BC+PQ)}{2} \cdot h = \frac{5}{4}a \cdot h$

The distance between BC and AD will be $\sqrt{3}a$, that is, the length of diagonal BF .

The distance between BC and AD will be $\frac{\sqrt{3}a}{2}$

And the height of the trapezium $BCQP$ will be $\frac{\sqrt{3}a}{4}$

$$\text{Area of trapezium} = \frac{5}{4}a \cdot \frac{\sqrt{3}a}{4} = \frac{5\sqrt{3}a^2}{16}$$

$$\text{Area of hexagon } ABCDEF = \frac{3\sqrt{3}}{2}a^2$$

$$\text{We have to find the ratio} = \frac{5\sqrt{3}a^2}{16} : \frac{3\sqrt{3}}{2}a^2 = 5:24$$

[VIDEO SOLUTION](#)

16. 2

$$\text{Given expression: } (a-b)^2 + (a-c)^2 + (a-d)^2$$

The given expression has just the sum of squares of the terms. So, the minimum value is either zero or positive.

If we can make all the values equal, we can get zero. But since all the values are integers and the sum 46 is not divisible by 4, we can't make everything equal.

So, the nearest four values are 12, 11, 11, 12.

$$\text{With this, the minimum value is } (12-11)^2 + (12-11)^2 + (12-12)^2 = 2$$

[VIDEO SOLUTION](#)

17. A

Let Lakshmi's income = A and expenditure = B .

Let Meenakshi's income = C and expenditure = D .

From $B:D = 2:3$, take $B = 2k$ and $D = 3k$.

From $A:D = 6:7$, $A = \frac{6}{7}D = \frac{6}{7} \cdot 3k = \frac{18}{7}k$.

Lakshmi's saving = $A - B = \frac{18}{7}k - 2k = \frac{4}{7}k$.

Let Meenakshi's saving = $C - D = C - 3k$. Given the savings ratio $\frac{\frac{4k}{7}}{C-3k} = \frac{4}{9}$.

Solve: $\frac{4}{7}k \cdot \frac{9}{4} = C - 3k \Rightarrow \frac{9}{7}k = C - 3k \Rightarrow C = \frac{30}{7}k$

So incomes ratio $A : C = \frac{18}{7}k : \frac{30}{7}k = 18 : 30 = 3 : 5$.

 VIDEO SOLUTION

18. **B**

Let the first term be a and the common ratio be r .

Given

$$a + ar + ar^2 = 52 \quad \text{and} \quad a^2r + a^2r^2 + a^2r^3 = 624.$$

From the first equation $a(1 + r + r^2) = 52$, so $a = \frac{52}{1 + r + r^2}$.

Substitute into the second:

$$\frac{52^2}{(1+r+r^2)^2} * (r + r^2 + r^3) = 624$$

$$\frac{r}{(1+r+r^2)} = \frac{3}{13}$$

$$3r^2 - 10r + 3 = 0$$

$$(3r - 1)(r - 3) = 0$$

Simplify to get an equation in (r) ; its real solutions are $r = \frac{1}{3}$ and $r = 3$. For an infinite geometric progression, we must have $(|r| < 1)$, so $r = \frac{1}{3}$.

Now

$$a = \frac{52}{1 + \frac{1}{3} + \frac{1}{9}} = \frac{52}{\frac{13}{9}} = 36$$

The sum of the infinite progression is

$$S = \frac{a}{1-r} = \frac{36}{1-\frac{1}{3}} = \frac{36}{\frac{2}{3}} = 54.$$

 VIDEO SOLUTION

19. **A**

Let coffee price = C Rs/kg and cocoa price = K Rs/kg.

From the two given mixtures:

$$0.16C + 0.84K = 240$$

$$0.36C + 0.64K = 320$$

Multiply both equations by 100 to remove decimals:

$$16C + 84K = 24000$$

$$36C + 64K = 32000$$

Subtract the first from the second:

$$(36C + 64K) - (16C + 84K) = 32000 - 24000$$

$$20C - 20K = 8000 \implies C - K = 400$$

Put $C = K + 400$ into $16C + 84K = 24000$:

$$16(K + 400) + 84K = 24000$$

$$16K + 6400 + 84K = 24000$$

$$100K = 17600 \implies K = 176$$

$$\text{So } C = 176 + 400 = 576 \text{ (Rs/kg).}$$

For the new mixture priced at Rs 376/kg, let the coffee fraction be p . Then

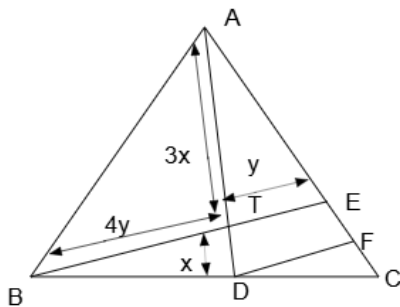
$$p \cdot 576 + (1 - p) \cdot 176 = 376$$

$$\text{Upon solving } p = \frac{1}{2}$$

Thus, coffee is (50%) of the new mixture. In 10 kg of this mixture, coffee = $10 \times 0.5 = 5\text{kg}$

[VIDEO SOLUTION](#)

20. B



In the question, it is given $AD : AT = 4 : 3$

$$\text{So, } TD : AT = 1 : 3$$

$$\text{Let, } TD = x, AT = 3x$$

$$\text{Also, } BE : BT = 5 : 4$$

$$\text{So, } BT : TE = 4 : 1$$

$$\text{Let, } BT = 4y, TE = y$$

Given, DF is parallel to BE

So, DF is parallel to TE

So, $\triangle ATE \triangle ADF$ (by A.A., as $\angle ATE = \angle ADF$ and $\angle AET = \angle AFD$)

$$\text{So, } \frac{AT}{AD} = \frac{TE}{DF}$$

$$\text{So, } \frac{3x}{x + 3x} = \frac{y}{DF}$$

$$\text{So, } DF = \frac{4y}{3}$$

Also, since DF is parallel to BE ,

So, $\triangle CFD \triangle CBE$ (by A.A., as $\angle CFD = \angle CEB$ and $\angle CDF = \angle CBE$)

$$\text{So, } \frac{CD}{BC} = \frac{DF}{BE}$$

$$\text{So, } \frac{CD}{BC} = \frac{\frac{4y}{3}}{5y} = \frac{4y}{15y} = \frac{4}{15}$$

$$\text{So, } \frac{CD}{BD} = \frac{4}{11}$$

$$\text{So, } BD : CD = 11 : 4$$

 VIDEO SOLUTION

21. 340

Let Chandan's one-day work = x .

Then Bipin = $2x$ and Ankita = $4x$.

All three work together for the first 20 days. Their combined daily work = $x + 2x + 4x = 7x$.

Work done in 20 days = $20 \times 7x = 140x$.

Bipin leaves after 20 days; Ankita and Chandan continue for the remaining $(60-20=40)$ days.

Their combined daily work = $4x + x = 5x$. Work done in 40 days = $40 \times 5x = 200x$.

Total work = $140x + 200x = 340x$.

We know that Chandan does x work in a day. So, the number of days it takes him to finish the work is 340.

Answer: 340 days.

 VIDEO SOLUTION

22. C

Let the total amount be 100 units (taking 100 makes percentage calculations easy).

Pinu receives 20% of the total, so Pinu gets 20 units.

The amount left after giving Pinu his share is $100 - 20 = 80$ units.

Meena receives 40% of this remaining amount, so Meena gets 40% of 80 = 32 units.

Now, the amount left for Rinu and Seema together is $80 - 32 = 48$ units.

Seema receives 20% less than Pinu. Since Pinu gets 20 units, Seema gets

$20 - 20\% \text{ of } 20 = 20 - 4 = 16$ units.

Out of the 48 units left for Rinu and Seema, Seema's share is 16 units, so Rinu gets

$48 - 16 = 32$ units.

Therefore, Pinu receives 20 units, and Rinu gets 32 units.

The required ratio of the amounts received by Pinu : Rinu = $20 : 32 = 5 : 8$.

 VIDEO SOLUTION